

REGINA, SK, Canada

WMO: 715140

Lat: 50.4269N

Lon: 104.6703W

Elev: 1894

StdP: 13.72

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.9	-21.9	-35.5	0.8	-26.9	-29.7	1.1	-21.5	34.9	13.5	30.7	13.1	8.4	290	0.757

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.6	87.7	66.5	83.9	65.1	80.7	63.7	70.5	81.5	68.0	79.5	65.8	77.0	12.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.7	105.4	76.7	64.0	95.6	73.7	61.6	87.8	71.5	35.7	81.2	33.5	79.4	31.7	77.0	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.4	25.8	23.1	-34.1	94.4	5.2	3.4	-37.8	96.9	-40.9	98.9	-43.8	100.8	-47.5	103.3
			-34.3	75.1	5.1	3.0	-38.0	77.3	-41.0	79.0	-43.9	80.6	-47.6	82.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	37.4	5.8	8.7	21.7	39.1	51.6	60.6	65.9	64.2	54.8	40.0	23.6	10.5	(6)
(7)		DBStd	24.15	15.16	14.35	14.27	10.60	8.47	6.50	5.64	6.40	8.44	9.32	12.50	14.22	(7)
(8)		HDD50	6229	1370	1157	878	347	84	3	0	0	45	329	793	1223	(8)
(9)		HDD65	10296	1835	1577	1343	778	421	163	59	94	321	774	1243	1688	(9)
(10)		CDD50	1611	0	0	0	19	132	320	492	440	188	20	0	0	(10)
(11)		CDD65	202	0	0	0	0	5	30	86	68	13	0	0	0	(11)
(12)		CDH74	2869	0	0	0	12	166	406	1008	966	301	10	0	0	(12)
(13)		CDH80	855	0	0	0	2	34	101	300	327	90	1	0	0	(13)
(14)	Wind	WSAvg	11.3	11.7	11.2	12.2	12.5	12.4	11.0	9.9	9.9	10.8	11.5	11.3	11.3	(14)
(15)	Precipitation	PrecAvg	15.6	0.6	0.5	0.7	0.9	2.1	3.0	2.5	1.7	1.3	0.9	0.6	0.7	(15)
(16)		PrecMax	23.1	1.6	1.3	2.2	2.5	4.6	8.0	7.2	4.6	3.5	4.2	1.5	1.4	(16)
(17)		PrecMin	8.8	0.0	0.0	0.0	0.0	0.4	0.6	0.3	0.2	0.1	0.0	0.0	0.2	(17)
(18)		PrecStd	3.3	0.3	0.3	0.5	0.6	1.1	1.8	1.6	1.1	0.8	0.8	0.4	0.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.7	40.2	57.9	74.5	84.6	88.7	91.8	92.8	88.0	74.7	58.7	42.8	(19)
(20)		MCWB	34.9	35.8	45.9	53.5	59.4	66.6	70.0	67.3	63.2	55.7	46.8	36.7		(20)
(21)		2%	DB	35.4	35.5	49.4	68.0	78.8	82.4	86.8	87.7	82.4	66.7	49.4	37.5	(21)
(22)		MCWB	32.8	33.1	42.1	50.7	57.3	63.9	68.8	66.0	60.9	52.2	41.8	34.1		(22)
(23)		5%	DB	31.7	32.3	42.8	62.6	73.7	78.6	83.0	83.4	76.6	60.2	43.4	33.7	(23)
(24)		MCWB	29.6	30.8	37.9	47.9	54.6	62.2	67.1	64.8	58.5	48.8	37.9	31.4		(24)
(25)		10%	DB	26.8	28.8	37.9	57.1	69.3	74.8	79.8	79.4	71.5	55.1	39.0	29.4	(25)
(26)		MCWB	25.4	27.3	34.9	45.6	52.7	60.1	65.7	63.2	56.8	46.3	35.2	27.6		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.3	36.5	47.1	55.4	62.3	71.0	75.9	72.1	66.0	57.4	47.5	37.2	(27)
(28)		MCDWB	37.9	39.4	56.0	71.6	76.3	83.4	85.5	83.6	81.1	71.0	57.9	41.8		(28)
(29)		2%	WB	32.9	33.2	42.2	52.1	58.8	66.8	71.9	69.1	62.8	53.3	42.2	34.2	(29)
(30)		MCDWB	35.3	35.0	49.1	65.6	73.9	77.5	82.4	81.8	77.0	64.5	48.7	37.2		(30)
(31)		5%	WB	29.6	30.8	38.1	49.1	56.6	64.1	69.1	66.8	60.2	49.8	38.3	31.2	(31)
(32)		MCDWB	31.7	32.4	42.5	60.5	70.3	74.5	79.3	79.6	74.0	58.8	42.6	33.6		(32)
(33)		10%	WB	25.5	27.2	35.0	46.1	54.4	61.7	66.8	64.7	57.5	46.9	35.4	27.6	(33)
(34)		MCDWB	26.9	28.7	37.8	55.9	66.7	71.7	77.4	76.5	69.3	54.3	38.8	29.4		(34)
(35)	Mean Daily Temperature Range	MDBR	17.8	18.0	17.4	22.0	24.8	21.7	23.6	25.3	25.1	21.6	17.9	16.7		(35)
(36)		5% DB	MCDBR	20.3	19.2	22.0	32.4	32.8	28.4	28.7	31.9	34.6	31.8	25.0	21.0	(36)
(37)		MCWBR	18.5	17.0	15.3	18.1	16.3	13.9	14.5	14.5	16.9	19.1	17.8	17.9		(37)
(38)		5% WB	MCDBR	20.1	18.5	20.8	30.2	27.6	23.4	25.1	27.1	30.0	29.3	23.2	20.2	(38)
(39)		MCWBR	18.5	16.5	15.1	17.9	14.7	12.9	14.2	14.0	15.5	18.1	16.8	17.5		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.239	0.312	0.272	0.327	0.345	0.350	0.359	0.368	0.325	0.279	0.249	0.239		(40)
(41)		taud	2.381	2.150	2.507	2.395	2.395	2.424	2.414	2.369	2.465	2.564	2.516	2.404		(41)
(42)		Ebn at Noon	263	256	301	291	289	287	283	274	277	275	260	246		(42)
(43)		Edh at Noon	19	32	27	34	36	35	35	35	28	21	17	16		(43)
(44)	All-Sky Solar Radiation	RadAvg	402	740	1166	1528	1846	1928	2082	1651	1229	735	402	284		(44)
(45)		RadStd	50	65	99	120	155	127	126	111	106	93	58	36		(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46)
(47)	Regional (15 neighbors)	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(47)

Nomenclature: See separate page